

Object Oriented Programming

Page No.:

Date:

YOUVA
PELLAR

```
String name = "Aditya";
```

```
int age = 28;
```

```
int rollNo = 101;
```

```
String college = "IIT Guwahati";
```

```
void (String name, ..... ) {  
    cout(.....);  
}
```

```
String name2 = "Rohit";
```

```
int age2 = 27;
```

```
int rollNo2 = 212;
```

```
String college2 = "IIT Madras";
```

Problem: - Need a lot of independent variables

- If we pass the student to a func" I need to pass 4 variables

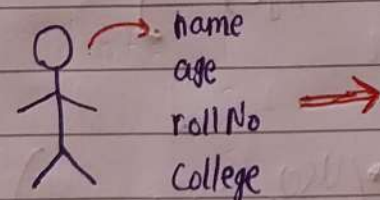
- we need to create new set of variables for new student

- No authority over data.

to solve these problems → oops was created

OOPs → different way of writing code

* Object Oriented Programming

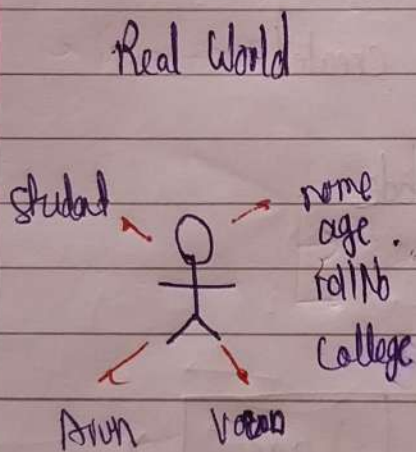
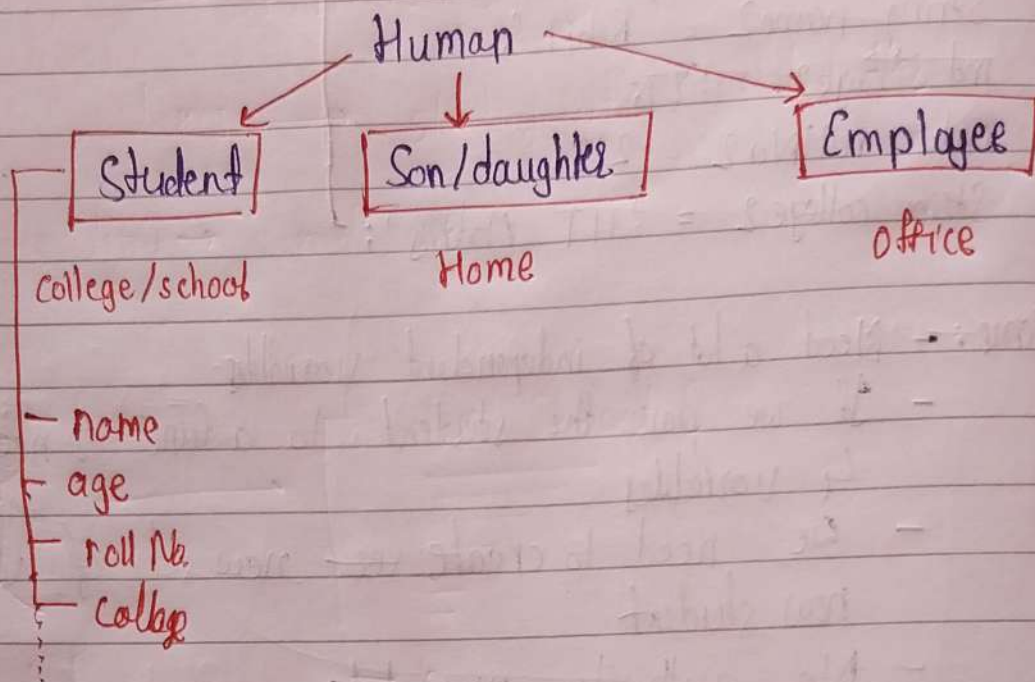
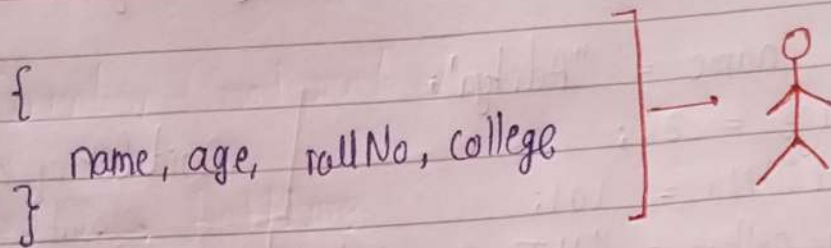


student

Real World

name	age
rollNo	college

programming world



Programming:

```
class Student {
    String name;
    int age;
    int rollNo;
    String college;
}
```

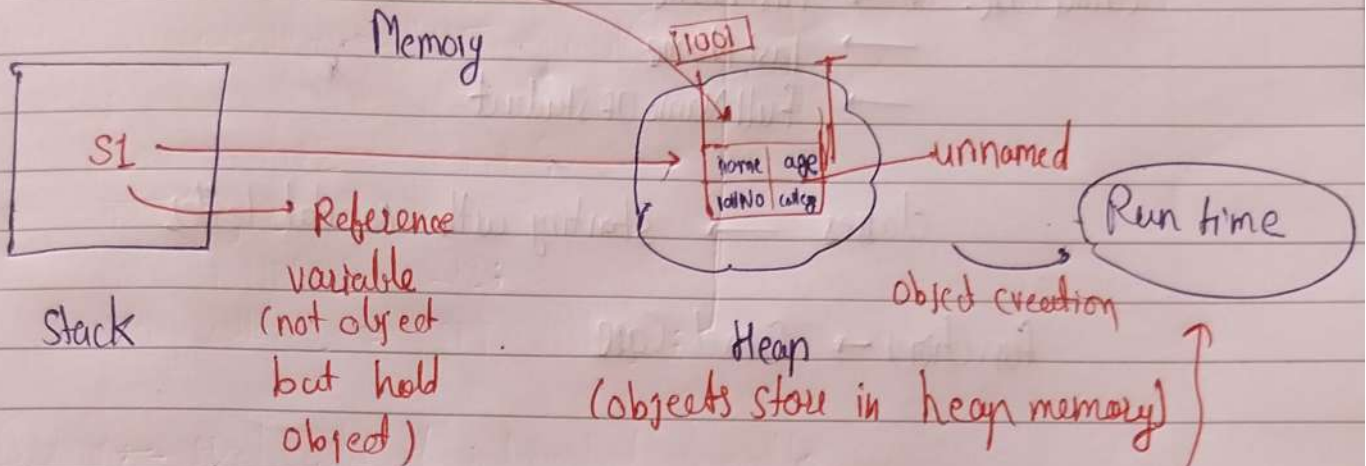
Blueprint of an object

Objects → data type → user
non primitive → defined data type

// Creating an object of class

Student s1;

s1 = new Student();



Student s1 = new Student();

compile time (static)

run time (dynamic)

s1.name = "Aditya";

s1.age = 28;

s1.rollNo = 101;

s1.college = "IIT Guwahati";

Student s2 = new Student();

s2.name = "Rohit";

s2.age = 27;

s2.rollNo = 102;

s2.college = "IIT Guwahati";

cout(s1.name); // Aditya

Naming Convention (Good practice)

Camel Case:
 eg: → First Name
 → last Name
 → Full Name OF Student

classes → starting with capital letter

Functions → Camel Case



Student
↳ Characteristic
↳ Behaviour

```
class Student {  
    _____  
    _____  
    _____  
    _____  
}
```

Animal

↳ characteristics (type, no. of legs, ...)
↳ Behaviour (make sound, eat, run, ...)

```
class Animal {  
    String type;  
    int no of legs;  
    ...  
    void makeSound () {  
        // code  
    }  
}
```

Functions

```
Animal cat = new Animal();  
cat.makeSound();
```



```
class Student {
```

```
    //
```

```
void print() {
```

```
    sout (name, age, ... );
```

```
}
```

```
}
```

```
Student s1 = new Student();
s1.print();
```

Real
Object

→ Digital
object

Animal

→ ✓

Student

→ ✓

Bank Account

→ ✓

location

→ ✓

→ long;

→ lat;

```
class BonAce {
```

```
    acc.No;
```

```
    ifsc-code;
```

```
    branch;
```

```
}
```